

The world of BGP (Border Gateway Protocol) and BGP - Hijacking



1. Internet Fundamentals

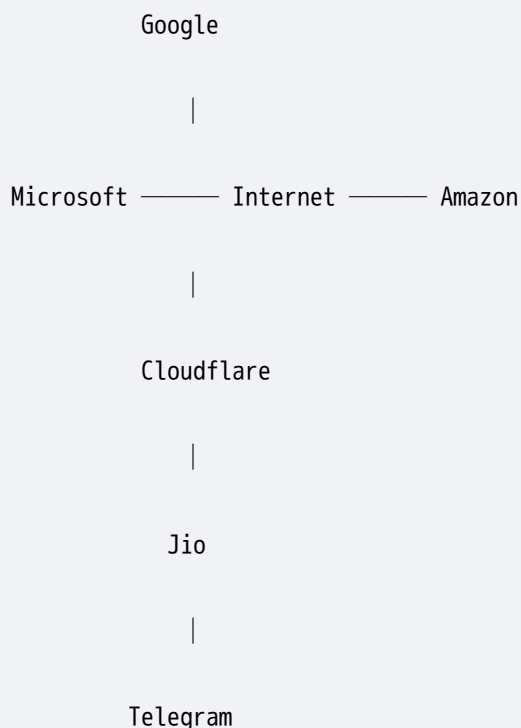
What is Internet?

Most people think Internet is one huge network.

Wrong.

Internet is actually a **Network of Networks**.

Example:



Each company owns its own network.

Together they form the Internet.

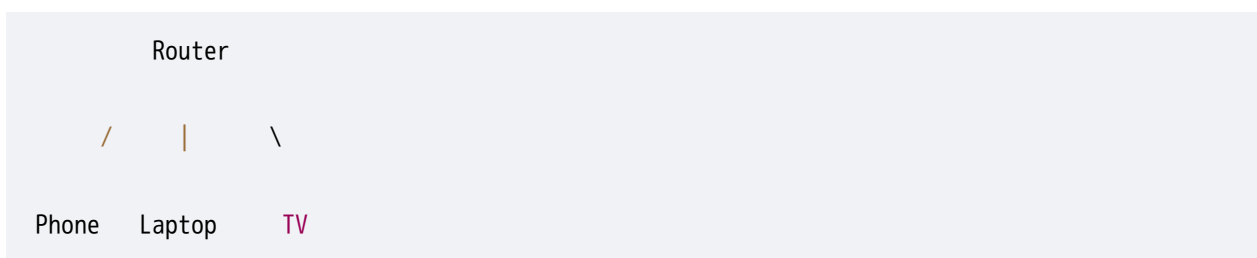
What is a Network?

A network is simply

| A group of connected devices that can communicate with each other.

Example

Home WiFi



Microsoft Network

Google Network

Telegram Network

All are networks.

What is a Router?

A Router is like a **traffic police**.

Its job is simple.

| Decide where the packet should go next.

It DOES NOT care about the final destination.

It only decides

| "What's the next stop?"

Example

Phone



Home Router



Jio



Google

What is a Packet?

When you send

Hello

Internet doesn't send one giant message.

It breaks it into smaller pieces.

These pieces are called **Packets**.

Packet 1

Packet 2

Packet 3

Destination joins them again.

What is an IP Address?

Every device on the Internet needs an address.

Just like

House Address

Every computer has

IP Address

Example

8.8.8.8

142.250.x.x

Without IP

Internet cannot deliver packets.

Why Prefixes?

Suppose Telegram owns

5000 IP addresses.

Instead of storing

149.154.160.1

149.154.160.2

149.154.160.3

...

Router stores

149.154.160.0/20

This is called a **Prefix**.

Much smaller.

Much faster.

Key Points

- ✓ Internet = Network of Networks
- ✓ Router forwards packets
- ✓ Every device has IP
- ✓ Routers store Prefixes instead of every IP



Chapter 2: Autonomous System (AS)

Why Do We Need Autonomous Systems?

Let's say there are only four companies in the world.

Google

Microsoft

Jio

Telegram

Each company owns:

- Thousands of Routers
- Millions of Servers
- Data Centers
- Fiber Cables

- Millions of IP Addresses

Now imagine your phone wants to open Telegram.

Should your ISP know **every Telegram server**?

For example:

149.154.160.1

149.154.160.2

149.154.160.3

...

Thousands more...

✗ No.

That would make the Internet impossible to manage.

So instead of treating every server separately...

The Internet treats the **entire Telegram network as ONE entity**.

This entity is called an **Autonomous System (AS)**.

Definition

An Autonomous System (AS) is a large network managed by a single organization that follows one routing policy.

Simple examples:

Company	Autonomous System
Google	Google Network

Microsoft	Microsoft Network
Jio	Jio Network
Telegram	Telegram Network

Think of an Autonomous System as **one big city**, even though it contains thousands of roads and buildings.

What is ASN?

Every Autonomous System gets a unique number.

This number is called the

ASN (Autonomous System Number)

Just like

- Aadhaar identifies a person.
- GST identifies a business.
- Roll Number identifies a student.

Similarly,

ASN identifies a network.

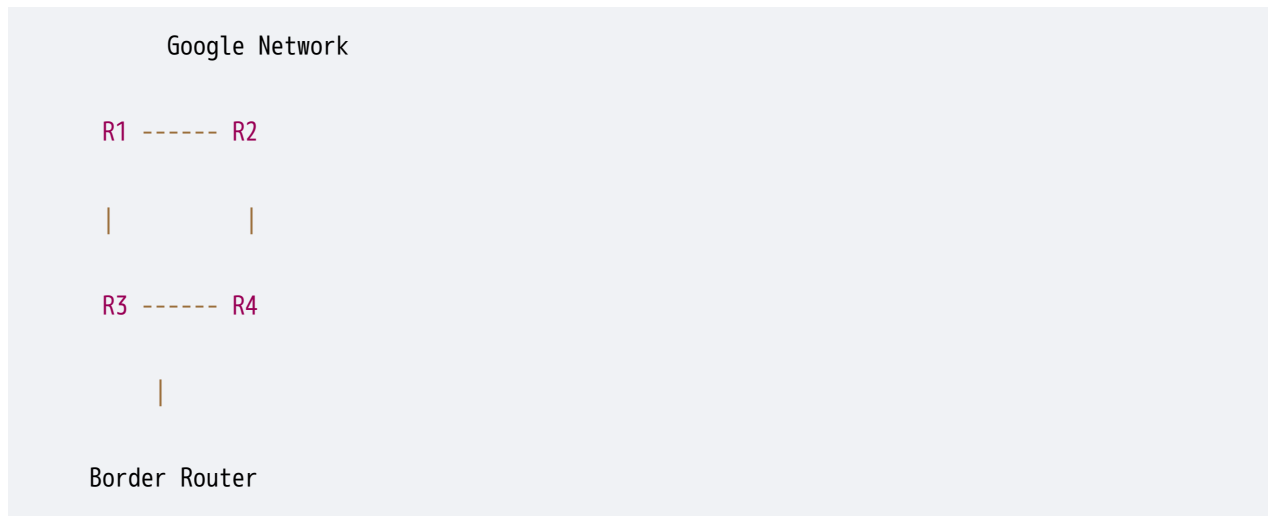
Real examples:

Company	ASN
Google	AS15169
Microsoft	AS8075
Cloudflare	AS13335
Amazon	AS16509

Important: Routers identify networks using **ASN**, not company names.

Internal Router vs Border Router

Imagine Google's network looks like this:



Google may have thousands of routers.

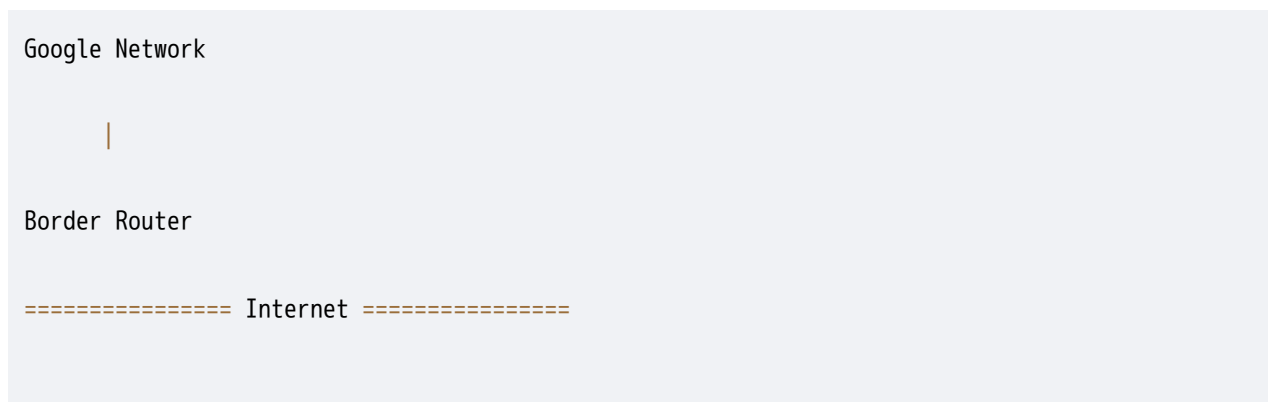
But only **one (or a few) routers** connect Google's network to the outside world.

These are called **Border Routers**.

Definition

A Border Router connects one Autonomous System to another Autonomous System.

Example:



Border Router



Jio Network

Think of it like an airport.

Inside an airport there are many counters, shops and gates.

But everyone enters through the **main entrance**.

Border Router is that entrance.

Why is it called "Border" Router?

Because it sits at the **border** of a network.

Example

Inside Google



Google Border Router



Outside World

Everything entering or leaving Google's network passes through the Border Router.

How do two companies communicate?

Now suppose Telegram wants to tell Jio:

"If anyone wants to reach my network, send the traffic to me."

Can Telegram send an email?

✗ No

WhatsApp?

✗ No

API Call?

✗ No

Instead,

their **Border Routers communicate using BGP.**

That's exactly why BGP exists.

We'll study BGP in the next chapter.

Real Life Analogy

Imagine four cities.

Delhi

Mumbai

Jaipur

Bangalore

Inside every city there are hundreds of roads.

But when you travel between cities,

you don't need to know every street.

You only need to know

"Which highway reaches that city?"

An **Autonomous System** is like a city.

BGP tells you **which highway reaches that city**.

Interview Questions

Q1. What is an Autonomous System?

A network managed by one organization under a common routing policy.

Q2. What is ASN?

A unique number assigned to every Autonomous System.

Q3. Why do we need Autonomous Systems?

Without AS, routers would have to remember millions of individual devices and routes, making Internet routing impossible to scale.

Q4. What is a Border Router?

A router that connects one Autonomous System to another Autonomous System.

Key Takeaways

- ✓ Internet is made of thousands of Autonomous Systems.
- ✓ Every Autonomous System has a unique ASN.
- ✓ Companies communicate through Border Routers.

✓ BGP works **between** Autonomous Systems.

Chapter 3: Border Gateway Protocol (BGP)

This is the heart of the Internet.

Why was BGP invented?

Suppose Telegram wants users from all over the world to reach its servers.

Question...

How will Jio know where Telegram is?

How will Airtel know?

How will Google know?

How will Microsoft know?

Somebody has to tell them.

That's exactly why **BGP** exists.

What is BGP?

Definition

BGP (Border Gateway Protocol) is the protocol used by Autonomous Systems to exchange routing information.

Simple English:

BGP tells one network how to reach another network.

Example



Telegram tells Jio,

"If someone wants to reach my network,
send traffic to me."

Jio stores that information.

That's BGP.

Breaking the Name

Border

Communication happens between different Autonomous Systems.

Example



Different networks.

Different borders.

Gateway

Gateway means

Entry or Exit point.

Border Router acts as the gateway.

Protocol

A protocol means

A set of communication rules.

Humans use

- Hindi
- English

Computers use

- HTTP
 - HTTPS
 - TCP
 - DNS
 - BGP
-

How does BGP actually work?

Suppose Telegram owns this IP Prefix

149.154.160.0/20

Telegram's Border Router sends a message to Jio.

Hello Jio 🖐️

149.154.160.0/20

belongs to me.

If anyone wants to reach [this](#) network,

send packets to me.

This message is called

Route Advertisement

Remember this term.

It is one of the most important concepts in BGP.

What information does BGP advertise?

BGP mainly advertises

- IP Prefix
- Next Hop
- AS Path

For beginners, just remember:

BGP advertises routes, not individual IP addresses.

Instead of saying

149.154.160.1

149.154.160.2

149.154.160.3

It simply says

149.154.160.0/20

Much smaller.

Much faster.

What is a BGP Session?

Before Telegram can send Route Advertisements,

both routers need to become neighbours.

This connection is called

BGP Session

Example

Telegram Border Router

Jio Border Router

Once connected,

they continuously exchange routing information.

Why does BGP use TCP?

BGP uses

TCP Port 179

Because routing information **must not be lost**.

Imagine Telegram tells Jio

"My network is here."

If that message gets lost,

Jio won't know where Telegram is.

TCP guarantees

✓ Reliable delivery

✓ Ordered delivery

✓ Error checking

That's why BGP uses TCP instead of UDP.

BGP Messages

Routers communicate using four important messages.

1. OPEN

Starts a BGP session.

Think of it as

"Hello, let's become neighbours."

2. KEEPALIVE

Sent regularly.

Purpose:

"I'm still alive."

If KEEPALIVE stops,
the neighbour is considered down.

3. UPDATE ★

Most important.

Used to

- Advertise new routes
- Withdraw old routes

Example

UPDATE

149.154.160.0/20

belongs to Telegram.

4. NOTIFICATION

Sent when an error occurs.

After sending this,

the BGP session closes.

Route Advertisement

Let's understand using a story.

Telegram opens a new Data Center.

Now Telegram wants everyone to know.

It sends

Telegram



Jio

Jio learns



Telegram owns this network.

Now Jio tells Airtel.

Jio



Airtel

Now Airtel tells Tata.

Airtel



Tata

Soon

the entire Internet knows.

This process is called

Route Propagation

Think of it like a rumor.

One person tells another.

Soon everyone knows.

Exactly how BGP updates spread.

What if Telegram changes its network?

Suppose Telegram shuts down one network.

Old Prefix

```
149.154.160.0/20
```

Now it should no longer be used.

Telegram sends another UPDATE.

```
Withdraw
```

```
149.154.160.0/20
```

This is called

Route Withdrawal

Routers remove that route from their tables.

Real Life Analogy

Imagine you own a restaurant.

You register on Zomato.

You tell them

My restaurant is on MG Road.

Now every customer can find you.

If tomorrow you shift,

you update your address.

Exactly the same happens in BGP.

Instead of restaurants,

it advertises networks.

Interview Questions

What is BGP?

A routing protocol used to exchange routing information between Autonomous Systems.

Why does BGP use TCP?

Because routing information must be delivered reliably.

It uses TCP Port 179.

What is Route Advertisement?

A BGP UPDATE message informing neighbours about reachable IP Prefixes.

What is Route Withdrawal?

A BGP UPDATE message informing neighbours that a previously advertised route is no longer valid.

Key Takeaways

- ✓ BGP connects different Autonomous Systems.
 - ✓ BGP exchanges routing information.
 - ✓ BGP uses TCP Port 179.
 - ✓ Routers first establish a BGP Session.
 - ✓ BGP advertises **Prefixes**, not individual IPs.
 - ✓ The two most important operations are:
 - Route Advertisement
 - Route Withdrawal
-



Next Chapter (Most Important)

Now we know **how routes are advertised**.

But another question comes.

Where does Jio store these routes?

How does Jio remember:

Google → This Way

Telegram → That Way

The answer is

Routing Table

And once you understand Routing Tables...

BGP Hijacking becomes almost obvious. 🔥



Chapter 4: Routing Table (The Brain of a Router)

Before learning BGP Hijacking, we need to answer one simple question.

When Jio receives a packet, how does it know where to send it?

The answer is... **Routing Table**.

What is a Routing Table?

A **Routing Table** is simply a table stored inside every router.

It tells the router:

"If you want to reach this network, send the packet here."

Think of it like **Google Maps**.

Google Maps stores routes.

Router stores routes too.

Example

Suppose Jio has learned routes from different networks.

Its routing table may look like this.

Destination Prefix	Next Hop
Google Network	Google Router
Microsoft Network	Microsoft Router
Telegram Network	Telegram Router
Cloudflare Network	Cloudflare Router

This table keeps growing as the router learns more routes.

What is "Destination Prefix"?

Remember Chapter 1?

Routers don't store every IP.

Instead of

```
149.154.160.1
149.154.160.2
149.154.160.3
...
```

They store

```
149.154.160.0/20
```

This is called the **Destination Prefix**.

What is "Next Hop"?

This is one of the most important concepts.

Suppose you want to go from Jaipur to Delhi.

Do you jump directly to Delhi?

No.

You first go to

Jaipur



NH48



Gurgaon



Delhi

Each step only knows

Where should I send you next?

Exactly the same happens on the Internet.

A router **doesn't need to know the complete path.**

It only needs to know

Who is the next router?

That router is called the **Next Hop.**

Example

Suppose Telegram has this prefix

149.154.160.0/20

Telegram advertises it to Jio.

Jio stores

Destination	Next Hop
149.154.160.0/20	Telegram Border Router

That's all.

Whenever a packet comes for Telegram,

Jio simply forwards it to Telegram's Border Router.

How does a packet travel?

Let's say you open Telegram.

Step 1

Your phone asks DNS

"What is Telegram's IP?"

DNS replies

149.154.167.91

Now your phone sends a packet.

Phone



Home Router



Jio

Jio checks its Routing Table.

149.154.167.91



Matches

149.154.160.0/20

Routing Table says

Next Hop



Telegram Router

Jio forwards the packet.

Phone



Home Router



Jio



Telegram Border Router



Telegram Server

Done.

How does Jio know that 149.154.167.91 belongs to 149.154.160.0/20?

This is done using

Longest Prefix Match (LPM)

Don't worry about binary calculations.

Just remember the idea.

Suppose the router has two routes.

Prefix
149.154.0.0/16
149.154.160.0/20

Destination IP

149.154.167.91

Both routes match.

Which one should the router choose?

It chooses the **more specific route**.

That is called

Longest Prefix Match.

Simple rule

More specific route wins.

Where do routes come from?

Question

Who fills the Routing Table?

Answer

BGP.

Whenever Telegram sends

149.154.160.0/20

Jio stores it.

Whenever Google advertises

its prefixes,

Jio stores those too.

So...

BGP keeps updating

the Routing Table.

Real Life Analogy

Imagine you're a courier company.

You have a notebook.

Delhi

↓

Truck A

Mumbai

↓

Truck B

Jaipur

↓

Truck C

Whenever a parcel arrives,

you open the notebook.

Destination = Delhi

↓

Truck A

Simple.

The notebook is the **Routing Table**.

Interview Questions

What is a Routing Table?

A Routing Table is a table inside a router that stores routes to different destination networks.

What is Next Hop?

The next router to which the current router forwards the packet.

Does a router know the complete path?

No.

A router usually knows only the best **next hop**.

What is Longest Prefix Match?

If multiple routes match a destination IP,

the router chooses the **most specific prefix**.

Key Takeaways

- ✓ Every router has a Routing Table.
- ✓ BGP fills the Routing Table.
- ✓ Routers store **Prefixes**, not every IP.
- ✓ Routers forward packets using the **Next Hop**.
- ✓ If multiple routes match, the **Longest Prefix Match** wins.



Chapter 5: BGP Hijacking

What is BGP Hijacking?

Imagine Jio's Routing Table looks like this.

Destination	Next Hop
Google	Google
Telegram	Telegram
Microsoft	Microsoft

Everything is working perfectly.

Now imagine...

An attacker sends a fake BGP UPDATE.

Hello Jio 🙋

149.154.160.0/20

belongs to me.

If Jio accepts this route...

The routing table changes.

Destination	Next Hop
Google	Google
Telegram	✗ Attacker
Microsoft	Microsoft

Nothing happened to Telegram.

Nothing happened to Jio.

Only **one route changed**.

This is called

BGP Hijacking.

Definition

BGP Hijacking is an attack where someone falsely advertises ownership of IP prefixes, causing Internet traffic to follow the wrong path.

Simple English

The attacker lies about the route.

He doesn't hack Telegram.

He doesn't hack your phone.

He lies to the Internet.

Why is BGP Hijacking possible?

Because the original Internet was built on **trust**.

Imagine you and your friends.

Your friend says

"Rahul shifted to Delhi."

You believe him.

Internet worked the same way.

If one network announced

"This IP belongs to me."

Other networks often trusted it.

Modern Internet has many protections, but this trust model is still one of the reasons hijacks can happen.

Normal Packet Flow

Suppose you open Telegram.

Phone



Home Router



Jio



Telegram



Telegram Server

Everything works.

During BGP Hijacking

Now attacker announces



Telegram is with me.

Packet flow becomes

Phone



Home Router



Jio



✗ Attacker



Telegram (maybe)

Notice something.

The attacker came **between** Jio and Telegram.

What can the attacker do?

There are mainly three possibilities.

1. Blackholing

The attacker simply drops all packets.

Phone



Jio



Attacker



Result

Telegram doesn't open.

Website looks down.

2. Forward the traffic

Instead of dropping packets...

The attacker forwards them.

Phone



Jio



Attacker



Telegram

Telegram still works.

Users may never notice.

But traffic passed through the attacker.

3. Redirect traffic

The attacker sends users somewhere else.

Example

Phone



Attacker



Fake Website

Very dangerous.

Real Life Analogy

Suppose your house address is

House 101
Green Street

Courier company has this address.

Everything works.

Now someone changes Google Maps.

House 101



House 999

Tomorrow

Amazon sends your parcel.

Courier reaches

House 999.

Did anyone steal your house?

No.

Did anyone hack Amazon?

No.

Only

the route changed.

That's exactly what BGP Hijacking is.

Real Incident (Pakistan vs YouTube)

One of the most famous examples happened in **2008**.

Pakistan wanted to block YouTube **inside Pakistan**.

To do this, a Pakistani ISP announced a route that effectively redirected YouTube traffic.

That incorrect BGP announcement accidentally spread beyond Pakistan.

As a result,

Users in many parts of the world couldn't reach YouTube for some time.

One wrong route...

Affected millions of users.

This is one of the classic examples of why BGP security matters.

Telegram Controversy

Recently,

Telegram CEO Pavel Durov alleged that an Indian telecom company used **BGP Hijacking** to disrupt Telegram traffic.

The telecom company denied the allegation.

Important

As of now,

there is **no public evidence proving intentional BGP Hijacking.**

So remember the difference.

✓ **BGP Hijacking is a real networking attack.**

✗ The specific Telegram case is still an allegation and not an established fact.

How do we prevent BGP Hijacking?

Today the Internet is much safer than before.

Some protections are:

RPKI

Allows routers to verify

whether a network is actually allowed to advertise a prefix.

Think of it as

Digital Identity Card for IP Prefixes.

Prefix Filtering

ISPs reject suspicious routes.

Route Monitoring

Companies like Cloudflare continuously monitor

whether someone suddenly starts advertising strange routes.

Interview Questions

What is BGP Hijacking?

A false BGP route advertisement that redirects Internet traffic.

Does the attacker hack the server?

No.

Usually the attacker changes the route,

not the server.

Why is BGP Hijacking possible?

Because BGP originally trusted route advertisements from neighboring networks.

Name one famous BGP Hijacking incident.

Pakistan Telecom accidentally affecting YouTube in 2008.

Key Takeaways

✓ BGP Hijacking = Fake Route Advertisement

✓ Server is usually **not hacked**

✓ Route is changed

✓ Traffic follows the wrong path

✓ Can cause outages, delays or interception

✓ RPKI helps reduce such attacks



Chapter 6: How Does BGP Choose the Best Route?

So far we learned:

- Networks advertise routes.
- Routers store them in a Routing Table.

But here's a big question.

What if Jio receives TWO routes for the same destination?

Example

Telegram advertises

149.154.160.0/20

through Airtel.

At the same time,

Telegram also advertises the same route through Tata.

Now Jio has two possible paths.

Path 1

Jio

↓

Airtel

↓

Telegram

Path 2

Jio

↓

Tata

↓

Telegram

Which one should Jio choose?

BGP has a **Best Path Selection Algorithm**.

Think of it like Google Maps.

If there are three roads to Delhi,

Google Maps chooses the best one.

Similarly,

BGP chooses the best route.

1. Local Preference (Highest Wins)

Local Preference tells your own network

which route it prefers.

Example

Via Airtel

Local Preference = 200

Via Tata

Local Preference = 100

Router chooses

Airtel

because **higher Local Preference wins**.

Simple memory trick

Local Preference = My Favorite Route

2. AS_PATH (Shortest Wins)

Every BGP UPDATE carries the list of Autonomous Systems it passed through.

Example

Path A

Jio

↓

Airtel

↓

Telegram

AS Path

55836

9498

62041

Length = 3

Path B

Jio

↓

Tata

↓

Cloudflare

↓

Telegram

AS Path

55836

4755

13335

62041

Length = 4

Which one is shorter?

Path A.

So router prefers Path A.

Memory Trick

Smaller AS Path = Better Route

Why does AS_PATH exist?

It has two purposes.

1. Choose shorter routes

2. Prevent routing loops

Suppose a router receives

AS Path

100

200

300

100

It notices

"My own ASN already exists."

That means

the route has come back to me.

Router immediately rejects it.

This prevents loops.

3. MED (Multi Exit Discriminator)

Suppose Google has two connections to Jio.

Google can tell Jio

"Please enter through this connection."

This suggestion is called

MED.

Smaller MED is preferred.

Think of MED as

Preferred Entrance Gate

4. Longest Prefix Match

We already discussed this.

Suppose router has

149.154.0.0/16

and

149.154.160.0/20

Destination IP

149.154.167.91

Both match.

Router chooses

149.154.160.0/20

because it is more specific.

Memory Trick

Route Advertisement vs Route Leak vs Route Hijack

Many beginners confuse these.

Let's simplify.

Route Advertisement

Normal.

Telegram tells everyone

149.154.160.0/20

belongs to me.

Everything is fine.

Route Leak

Accidental mistake.

Example

Company A accidentally advertises routes

that it shouldn't.

This can redirect traffic.

Usually

NOT malicious.

Route Hijacking

Malicious (or sometimes accidental).

Someone falsely claims

Telegram

belongs to me.

Now traffic comes to attacker.

Memory Table

Term	Meaning
Advertisement	Normal
Leak	Accidental
Hijack	False Route

How Does RPKI Prevent Hijacking?

Earlier,

Internet simply trusted everyone.

Today,

there is RPKI.

Full Form

Resource Public Key Infrastructure

Think of it as

Digital Aadhaar for IP Prefixes.

Suppose Telegram owns

149.154.160.0/20

Telegram creates a digital certificate saying

"Only my ASN can advertise this prefix."

Now attacker announces

149.154.160.0/20
belongs to me.

Router checks RPKI.

Certificate says

✗ Not Allowed.

Route rejected.

Simple.

Important Interview Differences

BGP vs OSPF

BGP	OSPF
Between organizations	Inside one organization
Internet Routing	Internal Routing

Path Vector	Link State
Slower	Faster
Huge Scale	Smaller Scale

Memory Trick

BGP = Between Companies

OSPF = Office Network

BGP Flow (One Minute Revision)

Telegram owns

149.154.160.0/20

↓

Border Router

↓

BGP UPDATE

↓

Neighbour Router

↓

Routing Table Updated

↓

Packet Arrives

↓

Router Checks Table

↓

Next Hop Selected

↓

Packet Forwarded

Complete Journey

Phone

↓

Home Router

↓

Jio

↓

Routing Table Lookup

↓

Telegram Border Router

↓

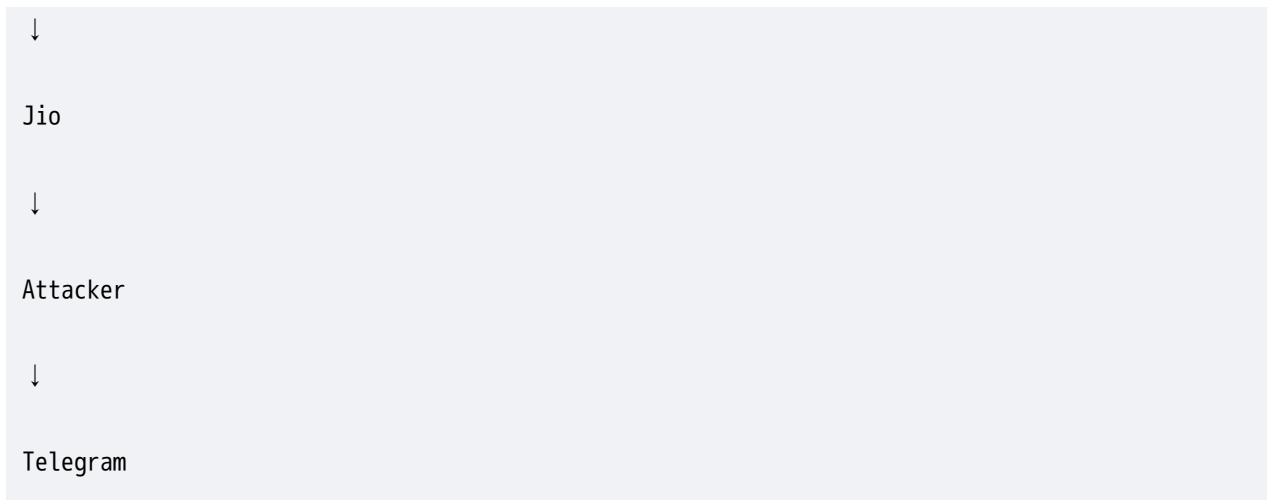
Telegram Server

During Hijacking

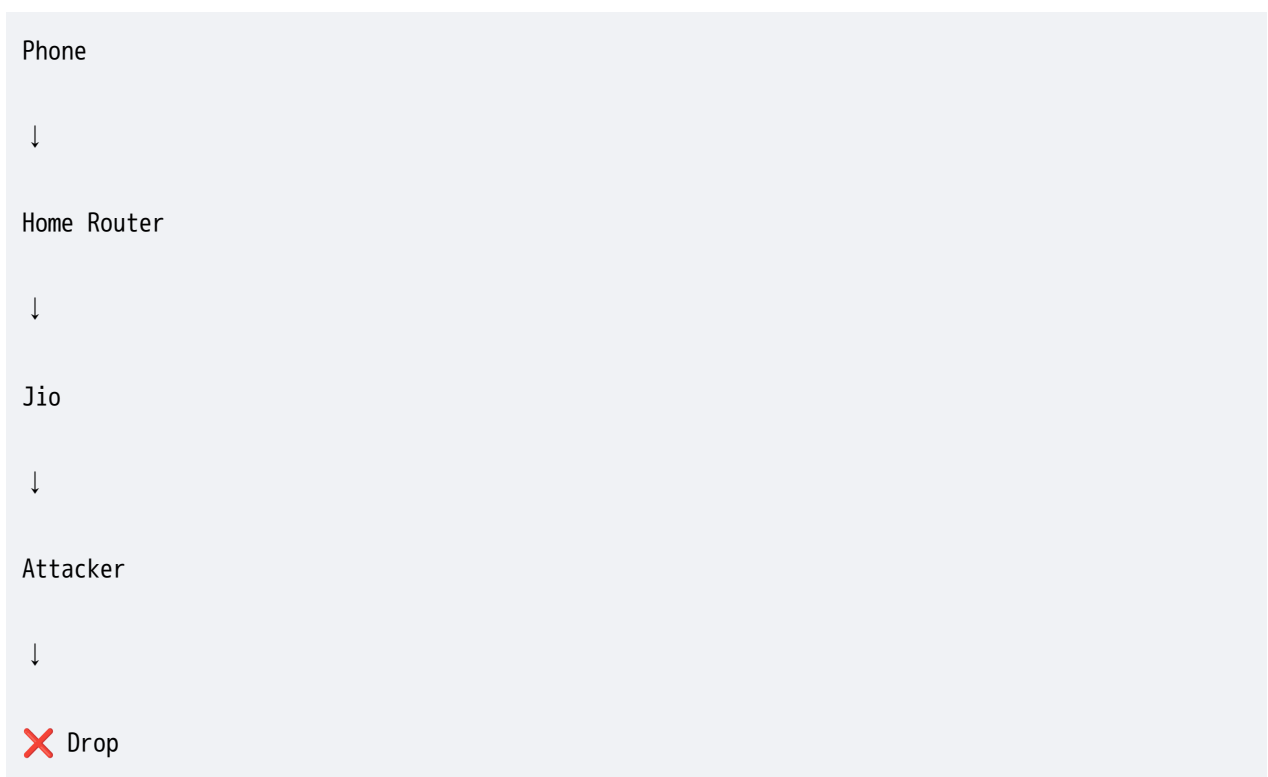
Phone

↓

Home Router



or



10 Things to Remember

- ✓ Internet = Network of Networks
- ✓ Every Network = Autonomous System
- ✓ Every AS has an ASN
- ✓ Border Routers run BGP

- ✓ BGP exchanges Routes
- ✓ Routes are stored in Routing Tables
- ✓ Routers forward packets using Next Hop
- ✓ BGP trusts route advertisements
- ✓ Fake Advertisement = BGP Hijacking
- ✓ RPKI helps verify advertisements